MTPro2 Math specimens

1 Three math sizes and compound expressions

Compare expressions in normal, superscript and nested superscript positions.

Inline example: $x^n + y^n = z^n$.

$$\sum_{i=1}^{n} \sin x + i^{\sin x} + i^{i^{\sin x}}$$

$$A,\ldots,Z \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$2^{A,\ldots,Z} \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$2^{2^{A,\ldots,Z}} \qquad a,\ldots,z \qquad \Gamma,\ldots,\Omega \qquad \Gamma,\ldots,\Omega \qquad \alpha,\ldots,\omega$$

$$\aleph_\alpha \times \aleph_\beta = \beta \iff \alpha \leq \beta$$

$$2^{\aleph_\alpha \times \aleph_\beta = \beta \iff \alpha \leq \beta}$$

$$2^{2^{\aleph_\alpha \times \aleph_\beta = \beta \iff \alpha \leq \beta}}$$

$$\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon$$

$$2^{\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon}$$

$$2^{2^{\forall \varepsilon > \alpha, \Gamma_\alpha \hookrightarrow \Gamma_\varepsilon}}$$

$$|x - a| < \delta \Longrightarrow |f(x) - l| < \varepsilon$$

$$2^{|x-a|<\delta \Longrightarrow |f(x)-l|<\varepsilon}$$

$$2^{2^{|x-a|<\delta \Longrightarrow |f(x)-l|<\varepsilon}}$$

$$\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}$$

$$2^{\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}}$$

$$2^{2^{\underbrace{V \times \cdots \times V}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \underbrace{V \times \cdots \times V}_{k+l}}}$$

$$\{x | x \neq x\} = \emptyset \qquad (A \cap B)^\circ \subset A^\circ \cap B^\circ$$

$$2^{\{x|x\neq x\}=\emptyset} \qquad (A\cap B)^\circ \subset A^\circ \cap B^\circ$$

$$2^{2^{\{x|x\neq x\}=\emptyset}} \qquad (A\cap B)^\circ \subset A^\circ \cap B^\circ$$

$$\omega = v + v(x, y)\, dx + w(x, y)\, dy + d\varkappa$$

$$2^{\omega=v+v(x,y)\,dx+w(x,y)\,dy+dx}$$

$$2^{2^{\omega=v+v(x,y)\,dx+w(x,y)\,dy+dx}}$$

$$d\omega = dv + \left(\frac{\partial w}{\partial x} - \frac{\partial v}{\partial y}\right) dx \wedge dy$$

$$2^{d\omega=dv+\left(\frac{\partial w}{\partial x}-\frac{\partial v}{\partial y}\right)dx\wedge dy}$$

$$2^{2^{d\omega=dv+\left(\frac{\partial w}{\partial x}-\frac{\partial v}{\partial y}\right)dx\wedge dy}}$$

$$\hat{x} + \widehat{X} + \widehat{xy} + \widehat{xyz} + \vec{A}$$

$$2^{\hat{x}+\widehat{X}+\widehat{xy}+\widehat{xyz}+\vec{A}}$$

$$2^{2^{\hat{x}+\widehat{X}+\widehat{xy}+\widehat{xyz}+\vec{A}}}$$

$$R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij}$$

$$2^{R_{ijkl}=-R_{jikl}=-R_{ijlk}=R_{klij}}$$

$$2^{2^{R_{ijkl}=-R_{jikl}=-R_{ijlk}=R_{klij}}}$$

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

$$2^{(f\circ g)'(x)=f'(g(x))\cdot g'(x)}$$

$$2^{2^{(f\circ g)'(x)=f'(g(x))\cdot g'(x)}}$$

$$f(x) = \begin{cases} |x| & x > a \\ -|x| & x \le a \end{cases}$$

$$2^{f(x)=\begin{cases} |x| & x>a \\ -|x| & x\le a \end{cases}}$$

$$2^{2^{f(x)=\begin{cases} |x| & x>a \\ -|x| & x\le a \end{cases}}}$$

$$\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx = \sqrt{\pi}$$

$$2^{\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx=\sqrt{\pi}}$$

$$2^{2^{\int_{-\infty}^{\infty} e^{-x\cdot x}\,dx=\sqrt{\pi}}}$$

$$X = \sum_i \xi^i \frac{\partial}{\partial x^i} + \sum_j x^j \frac{\partial}{\partial \dot{x}^j}$$

$$2^{X=\sum_i \xi^i \frac{\partial}{\partial x^i}+\sum_j x^j \frac{\partial}{\partial \dot{x}^j}}$$

$$2^{2^{X=\sum_i \xi^i \frac{\partial}{\partial x^i}+\sum_j x^j \frac{\partial}{\partial \dot{x}^j}}}$$

2 Standard accents

Compare ordinary accents with wide accents over longer expressions.

$$\mathring{x}$$

$$\dot{\Gamma} + \ddot{\Gamma}$$

$$\hat{A}, \quad \hat{M}$$
$$\widehat{A}, \quad \widehat{M}, \quad \widehat{xy}, \quad \widehat{xyz}, \quad \widehat{xyzw}, \quad \widehat{x+y+z+\cdots+w}$$
$$\tilde{A}, \quad \tilde{M}$$
$$\widetilde{A}, \quad \widetilde{M}, \quad \widetilde{xy}, \quad \widetilde{xyz}, \quad \widetilde{xyzw}, \quad \widetilde{x+y+z+\cdots+w}$$
$$\check{A}, \quad \check{M}$$
$$\widecheck{A}, \quad \widecheck{M}, \quad \widecheck{xy}, \quad \widecheck{xyz}, \quad \widecheck{xyzw}, \quad \widecheck{x+y+z+\cdots+w}$$

$$\bar{M}, \qquad \overline{M}, \qquad \overline{x+y+z}$$

3 Math alphabets

Compare alphabet styles and letters in subscript and superscript positions.

$$A_{\mathbf{X}}(f) = \mathbf{X(f)} = 2^{2^{\mathbf{X(g)}}}$$

ABGMXYZ
$ABGMXYZ$
ABGMXYZ
$\boldsymbol{ABGMXYZ}$
$\mathcal{ABGMXYZ}$
$\boldsymbol{\mathcal{ABGMXYZ}}$
$\mathfrak{ABGMXY3}$
$\boldsymbol{\mathfrak{ABGMXY3}}$
$\mathbb{ABGMXYZ}$

4 Parentheses at four matrix sizes

$$\hat{J}_x^{(1/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \hat{J}_x^{(3/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

$$\hat{J}_x^{(5/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{5} & 0 & 0 & 0 & 0 \\ \sqrt{5} & 0 & 2\sqrt{2} & 0 & 0 & 0 \\ 0 & 2\sqrt{2} & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 & 2\sqrt{2} & 0 & \sqrt{5} \\ 0 & 0 & 0 & 0 & \sqrt{5} & 0 \end{pmatrix}$$

$$\hat{J}_x^{(7/2)} = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ \sqrt{7} & 0 & 2\sqrt{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 2\sqrt{3} & 0 & \sqrt{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{15} & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & \sqrt{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{15} & 0 & 2\sqrt{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2\sqrt{3} & 0 & \sqrt{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{7} & 0 \end{pmatrix}$$

5 Formulas from physics and mathematics

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \qquad p_i = \frac{\partial L}{\partial \dot{q}_i},$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

$$\begin{cases} \nabla \cdot \boldsymbol{E} = \dfrac{\rho}{\varepsilon_0}, \\ \nabla \cdot \boldsymbol{B} = 0, \\ \nabla \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t}, \\ \nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{j} + \mu_0 \varepsilon_0 \dfrac{\partial \boldsymbol{E}}{\partial t}. \end{cases}$$

$$i\hbar\frac{\partial \psi}{\partial t} = \hat{H}\psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{r}, t)\right]\psi.$$

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{Z}, \qquad Z = \operatorname{tr}\left(e^{-\beta \hat{H}}\right), \qquad \beta = \frac{1}{k_{\mathrm{B}}T}.$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}\right) = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{f},$$

$$\nabla \cdot \boldsymbol{u} = 0.$$

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \quad \Longrightarrow \quad \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = \mathbb{P}\left(\bigcap_{m=1}^{\infty}\bigcup_{n=m}^{\infty} A_n\right) = 0.$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k}^2 = \begin{cases} (-1)^{n/2}\dbinom{n}{n/2}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$